



CAM-YOLO v8: A model for water meter word wheel region segmentation

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Abstract

The use of mobile phone APP meter reading reduces the probability of human intervention and misidentification, but due to the complex location and geographical conditions of the buried water meter, as well as the base code and gear in the water meter dial, there are problems of low accuracy and rough segmentation results when the water meter word wheel region is divided, so a YOLO v8 segmentation method based on coordinate attention mechanism is proposed. The backbone network incorporates the AM (Attention-MobileNet_block) module to apply the Coordinate Attention (CA) mechanism, which facilitates the extraction of multi-scale features, reduces model parameters and computational complexity, and enhances computing efficiency. In the feature fusion stage, the C2f_CA mechanism, based on the Bottleneck_CA module, effectively mitigates background interference while concentrating on the water meter's target region. Experimental results indicate that, compared to other segmentation models, our model achieves an F1-score of 98.61% and a mean Pixel Accuracy (mPA) of 99.11% for word wheel region segmentation, establishing a solid foundation for the precise identification of water meter code words in the future.

CCS Concepts

• Networks; • Network performance evaluation; • Network experimentation;

Keywords

semantic segmentation, CAM-YOLO v8 model, AM module, C2f_CA mechanism

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1 Introduction

The water meter word wheel region refers to the character area marked with the water meter reading [1] at a specific position on the dial, used to record the amount of water [2] and used for billing. Due to the complex embedding of the water meter and factors such as uneven illumination, stains, and moisture, the accuracy of the water meter word wheel region is low, resulting in significant errors. Numerous experts and researchers both domestically and internationally have conducted extensive innovative research and practical work on this: Jawas et al. [3] identified the digital sequence by analyzing the contour features of the water meter, improved the significant features in the image through manual segmentation of the water meter's digital sequence, and retrieved feature information using contour attributes. However, the operation was complex and the workload was large. Gu et al. [4] used NB-IoT technology to collect water meter images and uploaded the coordinate data of each character to the cloud platform using manual upright template images in the offline process. Local features were computed during the online processing, and the template features were compared with those of the input image to estimate the Perspective Transformation Matrix, and used a lightweight CNN network [5] to perform double-and-half character recognition of water meter character images, increasing labor costs. Chen et al. [6] used the improved U-Net network to perform binary classification, segmented the area of interest, and proposed an improved VGG16 network for reading recognition by using the Hough transform, vertical projection, and Gaussian filter for processing and single character segmentation, which laid the foundation for accurate recognition of code-words in the word wheel region in complex scenes. Liang et al. [7] suggested utilizing the original water meter dataset along with Faster R-CNN and SSD networks, and YOLO v3 object detection network to identify the area of interest and use the clustering algorithm to cut the target box dataset that meets the standard, which proves that the preprocessed dataset has better recognition effect. Hong et al. [8] proposed an end-to-end convolutional neural network localization and segmentation, using a key point positioning method based on spatial layout guidance to segment a single character, the premise is that the characters in the word wheel region are evenly spaced and aligned, using the pointer to determine the final reading of the water meter, and studied the impact of low-quality water meter images on water meter segmentation and reading. Zhao et al. [9] took the dataset of Hangzhou, China as the research object, and used deep learning technology to realize the detection and recognition of word wheel water meters, to improve the accuracy of water meter segmentation, at the same time, it also causes problems such

as large models, Many parameters, and high computing power overhead. Wang et al. [10] proposed an improved YOLO v5 [11] and GRU construction network to realize fast license plate recognition, which effectively solves the problem of using characterless segmentation in complex scenes, which improves the accuracy and speed, but there is a tilt phenomenon in the image of license plate recognition, which may produce a high error rate.

In order to address the aforementioned issues, this paper proposes a CAM-YOLO v8 (Coordinate-Attention-MobileNet_block-You Only Look Once v8) model for the water meter word wheel region segmentation and uses the dataset provided by the project partner enterprise. In contrast to the CSPDarknet-53 network utilized in YOLO v8, this study integrates an AM module into the backbone network. This module combines MobileNet V3 with an attention mechanism to capture more detailed and richer feature information, allowing for quick image inference and timely segmentation results.

In terms of feature fusion, the C2f_CA mechanism is used to selectively fuse feature maps of different sizes to improve the accuracy of the model in locating and segmenting water meter word wheel regions. To address the imbalance problem within and between classes, the Focal_EIOU [12] loss function was employed. The conclusion drawn through experimental comparison with other models is that using this model has relatively good accuracy and can effectively improve the accuracy and real-time performance of water meter word wheel segmentation, laying a good foundation for subsequent OCR.

2 Proposed Methods

YOLO v8 is based on the scaling series and provides different size model options at five scales, covering five models of different scales from YOLOv8n to YOLOv8x. Especially in the YOLOv8s version, it ensures high-precision output while maintaining fast processing. In addition, compared with the classical semantic segmentation model, such as U-Net, which is often used for multi-objective segmentation, cannot meet the requirements of accuracy and real-time, while the YOLO v8s model has more advantages in terms of speed and accuracy. After multiple experimental comparisons and combining the characteristics of two different datasets, an optimization model based on YOLO v8s is proposed, which takes the original image of $3 \times 640 \times 640$ as input, and outputs the segmentation mask of the specified category after processing by the back-bone network, feature fusion, and detection head, and the structure of the CAM-YOLO v8 model is as shown in Figure 1.

2.1 AM module

Inspired by MN-YOLOv5 [13], this paper proposes an AM module. This module is an improved version of MobileNet V3-small. Compared with the MobileNet V3-large version, the module is smaller in size and has higher computing efficiency in the process of deployment and inference, as shown in Figure 2. Firstly, the feature map passed through the conv_bn_hswish in Figure 1 is fed into the AM module, where a 1×1 convolution is first used to increase dimensionality, followed by batch normalization (BN) and a non-linear activation function to improve the model's ability to represent non-linear relationships. Next, a depth-wise separable convolution 3×3

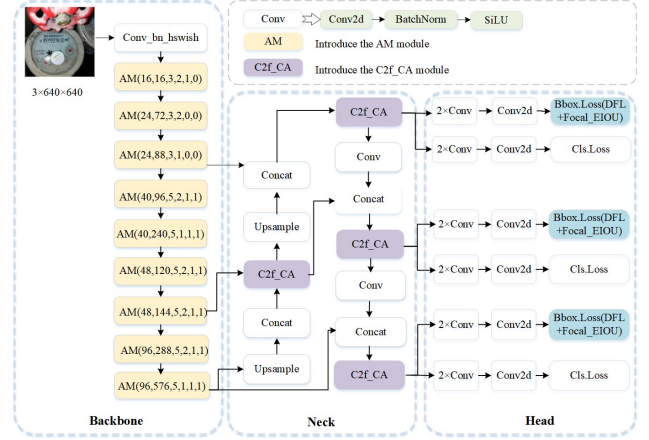


Figure 1: The structure of the CAM-YOLO v8 model.

is used for each channel, and a non-linear activation function is applied to further process the feature map. Improving the model's sensitivity to both spatial and channel information, this paper innovatively introduces the coordinate attention mechanism. The module integrates pooling, normalization, and non-linear activation functions in the X and Y directions, enabling more precise localization and identification of the target object. Ultimately, after applying 1×1 convolution for dimensionality reduction, the final output feature map is achieved by summing the input and output results when the number of input and output channels remains the same.

By introducing a series of AM modules, the features of the water meter word wheel region are extracted. In contrast to standard convolution in the backbone network of the YOLO v8 model, the use of depth separable convolutions significantly decreases the model's parameter count. The parameters of the AM module can effectively decrease the receptive field of each feature pixel, thereby lowering the model's computational complexity and improving segmentation performance.

2.2 Optimized the Coordinate Attention

The Coordinate Attention (CA) mechanism [14] embeds positional information into channel attention, enabling the acquisition of information from a larger area without additional overhead. Furthermore, to avoid compressing all spatial information into the channels, it decomposes channel attention into two parallel one-dimensional feature encodings to efficiently integrate spatial coordinate information into the generated attention map. This allows the model to capture long-range spatial relationships in one direction while preserving accurate positional information in the opposite direction.

This study has improved the coordinate attention mechanism, as illustrated in Figure 3. The detailed steps are below:

(1) The residual feature map $Y(H) \times X(W)$ is learned by the residuals, which helps the gradient flow better and prevents the gradient disappearing problem in the deep network. There are two different processing methods for the residual feature map: one is to directly weigh the residual feature map, and the other

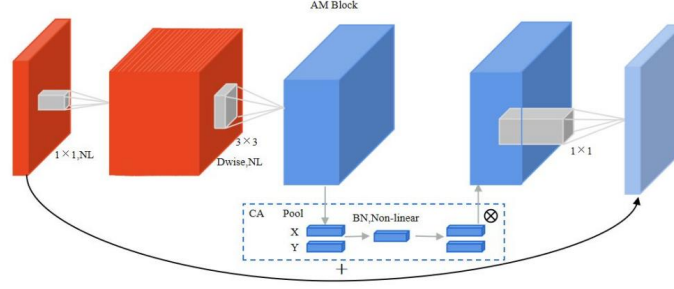


Figure 2: The structure of the AM module.

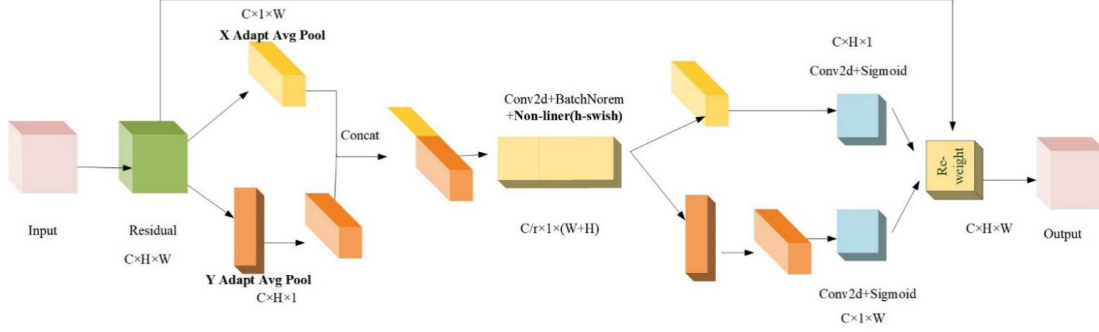


Figure 3: The structure of the optimized coordinate attention mechanism.

is the "bidirectional adaptive residual", which is performed in the following steps:

(a) In the operations along the X and Y axes, adaptive average pooling is applied to the intermediate feature tensors of the residual feature maps to perform two one-dimensional feature encodings. This involves using two pooling kernels (None,1) or (1,None) separately averaging each channel feature along the horizontal and vertical coordinates. X is used as an input, $X = [x_1, x_2, \dots, x_C] \in \mathbb{R}^{C \times H \times W}$, consisting of multiple feature maps, each of which can be expressed as x_i , indicates the number of channels and Y is the output, $Y = [y_1, y_2, \dots, y_C] \in \mathbb{R}^{C \times H \times W}$ is the same size as X, as shown in Equations 1 and 2;

(b) The feature map obtained in both directions ($[z^h, z^w]$) is concatenated along the spatial dimension, and its dimension is reduced to the original C/r , through the shared convolution kernel, and the feature map F_1 (reduced using the 1×1 convolutional kernel) is passed through the h-swish activation function to produce the feature map f , with r representing the scaling factor, as detailed in Equation 3. Additionally, The feature map f is divided to extract spatial information from the image, highlighting critical positional details. According to the original height and width, the convolution is carried out to obtain the same number of channels as the original feature plot F_h and F_w , and then combined with the Sigmoid activation function, the final attention weights g^h and g^w are obtained, as shown in Equations 4 and 5.

$$z_c^h(h) = \frac{1}{W} \sum_{0 \leq i < W} |x_c(h, i) \quad (1)$$

$$z_c^w(w) = \frac{1}{H} \sum_{0 \leq j < H} |x_c(j, w) \quad (2)$$

$$f = h - swish[m] \sum_{0 \leq i < W} x_c(h, i), m = F_1([z^h, z^w]) \quad (3)$$

$$g^h = \delta(F_h(f^h)) \quad (4)$$

$$g^w = \delta(F_w(f^w)) \quad (5)$$

(2) The resulting weights are combined with the direct weights to produce the final output feature map, as illustrated in Equation 6.

$$y_c(i, j) = x_c(i, j) \times g_c^h(i) \times g_c^w(j) \quad (6)$$

In summary, the mechanism employs adaptive average pooling to automatically adjust the input feature map size, providing flexibility. Additionally, the h-swish activation function effectively preserves higher-order information about the features, aiding the network in more accurately learning the complex relationships between them.

2.3 C2f_CA mechanism

The C2f_CA mechanism based on the Bottleneck_CA module can enhance feature fusion segmentation performance and reduce information redundancy during upsampling. The mechanism selectively fuses the resolution feature maps of different sizes, and realizes the adaptive weight adjustment between feature channels through the coordinate attention mechanism, emphasizing the key spatial areas, and improving the model's segmentation ability of the target features. Figure 4 (b) illustrates the C2f_CA mechanism, where the input feature map undergoes 1×1 convolution, Batch Normalization,

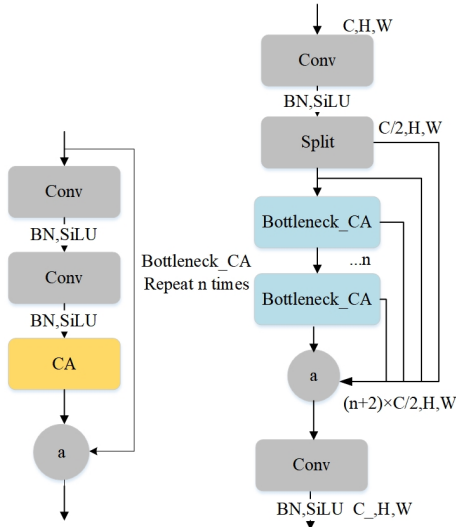


Figure 4: The structure of the Bottleneck_CA module (left (a)) and C2f_CA mechanism (right (b)).

and SiLU activation for better non-linear feature representation. It is then split into two segments, each with half the number of channels, maintaining spatial dimensions to enhance feature diversity. One part is processed through n Bottleneck_CA modules, while the other bypasses them and is concatenated with the processed features. This fusion improves feature representation by combining high-level features with low-level ones, resulting in a feature map with a channel size of $(n+2) \times C/2$. The final feature map is passed through convolutional layers to produce the output. Figure 4 (a) shows the Bottleneck_CA module, which processes the feature map through two convolutions, BN, and SiLU activation, followed by a coordinate attention mechanism to enhance cross-channel information interaction. The result is added to the input map only if the input and output channel numbers match [15].

3 Analysis of Experimental results

3.1 Experimental Datasets

The experimental environment of this article is deployed on Windows 11, equipped with the 13th generation Intel (R) Core (TM) i5-13600KF processor, NVIDIA GeForce RTX 4070 graphics card, and 32GB of memory. The deep learning framework used is PyTorch 2.0.1 with CUDA 11.8, Python 3.8.3, and Labelme 3.17.0. The water meter dataset is provided by project partners, consisting of 306 images. Annotate the Labelme using the water meter wheel area and enhance the data using imgaug to expand the dataset to 2142 images. Write a custom script to convert comments from json to the txt format required by the YOLO model. In this study, the annotated txt file and the training dataset in jpg format are fed into the CAM-YOLO v8 model for training. The dataset is randomly partitioned into training, validation, and test sets (6:2:2), the parameter is set to learning rate=0.0001, batch size= 4, momentum=0.937, weight_decay=0.0005, epoch=300, IOU=0.7.

3.2 Comparative experiments

As shown in the comparative results in Table 1, PSPNet performs slightly worse than both HrNet and DeepLabV3+ across all metrics. The HrNet and DeepLabV3+ models outperform PSPNet, achieving 97.68%, 97.65%, 97.66%, and 98.80% (HrNet), and 96.85%, 98.09%, 97.47%, and 99.01% (DeepLabV3+) in Precision, Recall, F1-score, and mPA, respectively. While HrNet shows better performance due to its complex structure and powerful feature extraction capabilities, DeepLabV3+ struggles with small targets, yielding suboptimal results. The CAM-YOLO v8 model, however, outperforms HrNet by 1.39%, 1.88%, 1.64%, and 0.31% in Precision, Recall, F1-score, and mPA, respectively, demonstrating superior segmentation accuracy in challenging scenarios. Additionally, compared to HrNet, the CAM-YOLO v8 model has fewer trainable parameters (6.86 MB) and lower FLOPs (30.10 GB), highlighting its superior segmentation performance and efficient inference speed.

Figure 5 shows a visualization of our model and other models performing word wheel region segmentation on this dataset. The segmentation results are shown in Figure 5, where subfigures (c–e) depict rough segmentation outcomes, exhibiting blurred boundaries or sawtooth phenomena in certain word wheel regions. In contrast, the segmentation result in Figure 5 (f), obtained by the proposed method, achieves a more precise segmentation effect, accurately predicting the water meter word wheel region and aligning well with the mask label. This shows that the method can more clearly divide the word wheel region of different types of water meters, the word wheel region is not disturbed by other dial elements, and the division is clearer and smoother, which reduces the sawtooth phenomena.

3.3 Ablation experiments

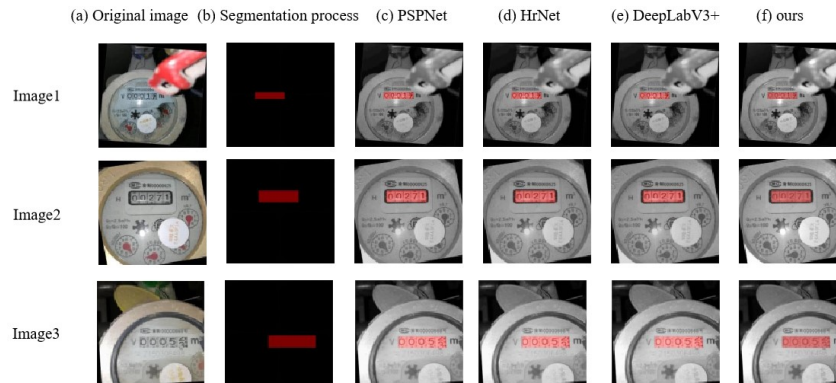
To assess the effectiveness of the experimental model, this paper gradually improves and compares the water meter dataset based on YOLO v8. The experimental outcomes, as shown in Table 2, demonstrating that the introduction of AM modules into the backbone network greatly improves feature extraction abilities, and the performance of the model can be improved from 94.67% to 97.71%, which makes the model performance significantly improved. Secondly, the C2f_CA module was implemented during feature fusion, resulting in a 0.91% increase in accuracy, which further enhanced the segmentation precision of the water meter word wheel region. Finally, to balance the sample distribution, the precision was increased by 0.45% with the addition of the Focal_EIOU loss function. These findings highlight the effectiveness of the proposed approach in feature extraction, cross-scale integration, and sample balancing.

4 Conclusions

To improve the accuracy of water meter word wheel region segmentation and reduce computational complexity, this paper proposes a semantic segmentation model based on YOLO v8, which combines AM module and attention mechanism to reduce model parameters and meet real-time requirements. The word wheel region can be accurately located, reducing other interfering elements and improving segmentation accuracy. In feature fusion, the C2f_CA mechanism is employed to accurately locate and focus on the word wheel region, reducing interference from elements on the dial and improving

Table 1: The segmentation results of the seven models

Model	Precision	Recall	F1-score	mPA	Parameters/10 ⁶	FLOPs/GB
PSPNet	96.53%	95.98%	96.25%	97.95%	2.38	9.42
HrNet	97.68%	97.65%	97.66%	98.80%	9.64	58.32
DeepLabV3+	96.85%	98.09%	97.47%	99.01%	5.81	41.30
ours	99.07%	99.53%	99.30%	99.11%	6.86	30.10

**Figure 5: The renderings of different types of water meters.****Table 2: The results of ablation experiments**

No	Model	Precision	Gain
0	YOLO v8	94.67%	-
1	YOLO v8+AM	97.71%	+3.04
2	YOLO v8+C2f_CA	96.60%	+1.93
3	YOLO v8+AM+C2f_CA	98.62%	+3.95
4	ours	99.07%	+4.40

the segmentation accuracy of the word wheel region boundary. Meanwhile, the dataset is expanded through data augmentation methods. The experimental results show that the CAM-YOLO v8 model has an accuracy of 99.07%, Recall rate of 99.53%, an F1-score of 99.30%, and mPA of 99.11% on the water meter dataset. The proposed method has high segmentation accuracy, and the edges of the water meter word wheel region are finer and smoother, providing support for future research to determine the water meter readings in the segmented water meter wheel region.

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